

Build Text Generator Model using Amazon SageMaker

Overview

Amazon sagemaker is a managed service on AWS cloud. It provides tools to deploy any Machine Learning algorithm on the cloud for predictive analysis. Data Scientist needs to deploy their built Machine Learning model in real-time, but they lack software engineering skills and rely on software engineers to deploy their model. On the other hand, Software engineers are not data scientists. Developing a full-scale ML service requires the developer to have a deeper understanding of the process, which is quite overwhelming in most cases. The gap between the data scientist and software engineer can be bridged by aiding a data scientist with the necessary software skills to deploy their machine learning model. This is where amazon sagemaker comes in very handy. With SageMaker, data scientists and developers can quickly and easily build and train machine learning models and directly deploy them into a production-ready hosted environment.

In this project, we will build a text generator model for amazon food reviews using the LSTM algorithm in PyTorch. The model will be trained and deployed on amazon sagemaker.

Aim

- To understand text generation
- To train and deploy the LSTM text generator model on amazon sagemaker

Tech Stack

- Language: Python
- Libraries: pytorch, pandas, numpy, regex,boto3

Data Description

The dataset contains food reviews extracted from different review sections. It has various attributes but only the review feature will be used to train the model.

Approach

- Data Loading
- Data Preprocessing

- Data cleaning
- Creating a sequence of data
- Tokenization
- Model Building and Training
 - LSTM model building in PyTorch
- Model Deployment on Amazon Sagemaker

Modular Code Overview

Input

|_reviews.csv

MLPipeline

|_Data_Processing.py

|_Model.py

|_Predict.py

|_Tokens.py

|_Train.py

NoteBook

|_Sage_Maker_Text_Generator.ipynb

Output

|_model.pt

Sagemaker_Deployment

|_Deployement_ipynb

|_Text_Generator_Development.ipynb

|_textgenerator_py

|_textgeneration.py

Engine.py

Readme.md

requirements.txt

Once you unzip the textgenerator_code.zip file, you can find the following folders within it.

1. Input
2. MLPipeline
3. NoteBook
4. Output

5. Sagemaker_Deployment
 6. Engine.py
 7. Readme.md
 8. requirements.txt
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1. The Input folder contains the data that we have for analysis. In our case, it includes reviews of the product
 2. The Notebook folder contains the jupyter notebook file of the project
 3. The ML_pipeline is a folder that contains all the functions put into different python files, which are appropriately named. These python functions are then called inside the Engine.py file
 4. The Output folder contains the saved model
 5. Sagemaker_Deployment folder contains the files for deployment
 6. The requirements.txt file has all the required libraries with respective versions. Kindly install the file by using the command **pip install -r requirements.txt**
 7. **All the instructions for running the code are present in Readme.md file**

Takeaways

1. What is a text generator?
2. PyTorch introduction
3. How does text generation work?
4. What are LSTM and its application?
5. What is a sagemaker?
6. Uses of sagemaker
7. How does sagemkaer work?
8. How to create a notebook instance in sagemaker?
9. How to upload data on the s3 bucket?
10. How to fetch data from the s3 bucket?
11. Data cleaning
12. How to create tokens from sentences?
13. How to convert tokens to integers?
14. Model training in sagemaker
15. Deploying model on sagemaker

